

CAPEX & WBS Control Hub

What was built — proof-of-concept build report



Atha Group · Capital expenditure and CWIP control · 6 August 2026

Executive summary

A working proof of concept has been built and runs locally. It gives Atha Group project-wise control over capital expenditure by monitoring approved budget against both committed cost (open purchase orders) and actual cost (vendor bills) — so available budget is visible before further procurement is approved, which is the control the client asked for.

The application is independent of Zoho ERP and integrates with it through a connector. Zoho remains the accounting and procurement system of record; the control layer, the multi-level work breakdown structure and the budget register live here, because the research established that Zoho ERP cannot carry them.

At a glance

Item	Delivered
Application	Runs at <code>http://127.0.0.1:8000</code> via <code>python app/run.py</code>
Stack	Python 3.11 · FastAPI · SQLite · zero-build front end. No npm, no external services.
Code	~2,850 lines across engine, API, connector and UI
Screens	17, all rendering; 17 API endpoints, all returning 200
Evidence base	1,527 researched claims, 1,402 citable after adversarial verification
Zoho grounding	869 operations across 78 modules, taken from Zoho's own OpenAPI specification
Demo dataset	2 entities · 2 plants · 3 projects · 3-level WBS · 10 budget heads · 21 PRs · 15 POs

The single most important property

For any purchase order, billed + open commitment = ordered. Billing moves value from commitment to actual and never adds to both, so a purchase order can never be counted twice. This is verified live in the running system: PO-2026-0003 shows ₹17,40,000 billed plus ₹17,00,000 open commitment against ₹34,40,000 ordered — and its exposure does not move as billing progresses.

How this was produced

The build was preceded by a structured research and verification programme, because the brief required every SAP and Zoho capability claim to be evidenced rather than assumed. That programme changed the design in ways that guesswork would not have:

Stage	Output	Why it mattered
Intake	107 client requirements extracted; 22 contradictions logged	Established that the client document contains no WBS at all — only project plus budget head
Research	15 parallel lanes, 1,527 evidenced claims	SAP Project System behaviour, Fiori patterns, Zoho ERP endpoints, licensing, Indian accounting

Stage	Output	Why it mattered
Verification	Cross-assigned adversarial re-checking of every claim	36 of 1,199 quote-checked claims failed string matching — a ~3% fabrication-or-drift rate
Specification	Zoho's official OpenAPI bundle downloaded and parsed	Replaced page-scraping with vendor ground truth: 869 operations, 163 OAuth scopes
Contract freeze	12 registries fixed before any code	Formulas, statuses, entities, tokens and messages could not drift during the build

The control logic

Everything in the application derives from three formulas, frozen before the build began and taken directly from the client's own document, whose figures were re-verified line by line. All thirteen numeric rows in that document are internally consistent; no arithmetic error exists in the source.

```

Exposure          = Open PO Commitment + Actual CWIP
Available Budget  = Current Approved Budget - Exposure
Utilisation %     = Exposure / Current Approved Budget

Current Approved Budget = Original Budget
                        + Approved Supplements
                        - Approved Returns
                        +/- Approved Transfers
  
```

The original approved budget is immutable. Supplements, returns and transfers are recorded as separate versioned lines, each carrying requestor, approver, date, reason and approval reference — so the original figure is never overwritten and the revision history is auditable.

Three controls, each answering a specific failure mode

1 • Anti-double-count

A purchase order line contributes to commitment only for its unbilled balance. When a bill arrives, value moves from commitment into actual cost; it is never added to both. Total exposure is therefore unchanged by billing progress, which is exactly what the client's own worked example requires.

Transaction position	PO value	Actual bill	Open commitment	Budget exposure
PO approved, not billed	₹20L	₹0	₹20L	₹20L
Partially billed	₹20L	₹12L	₹8L	₹20L
Fully billed	₹20L	₹20L	₹0	₹20L
Cancelled before billing	₹20L	₹0	₹0	₹0

2 • Anti-under-count

This control exists because verification of SAP documentation found something the obvious reading misses. Commitment relief is governed by two independent gates, not one: the goods-receipt indicator on the purchase order item, and the goods-receipt-non-valuation flag. Where they disagree — specifically when the GR indicator is set but the receipt is non-valuation — commitment is released at receipt while actual cost only arrives at invoice, so the value sits in neither bucket and consumption is understated.

How the application avoids it

Commitment is relieved only against billing, and received-not-billed is tracked and reported as an explicit exposure bucket. Nothing can fall between the two. The demo dataset carries ₹5,00,000 in this state so the behaviour is visible on screen.

3 • Budget-owner resolution

This one was found while testing the interface, not during design. Spending is posted on leaf WBS elements while budget is approved one or two levels above them. Checking a leaf's own budget line would reject every transaction for want of a budget that was never meant to sit there; checking only a parent's own line would let its budget look untouched while children consumed it. Both are ways an availability check silently fails.

The check now resolves to the nearest ancestor that actually carries budget and evaluates its rolled-up position. The message states this plainly rather than leaving the user to work it out:

Live message from the running application

“The proposed value of ₹20,00,000 exceeds the available budget of ₹19,10,000 for CAPEX-2026-001.03.01 (budget is held at CAPEX-2026-001.03) (Plant & Machinery) by ₹90,000. Submit a budget revision or request exception approval.”

Every message names the project, WBS element, budget head, available amount, proposed amount and shortfall, and offers a next action — the standard the brief set.

Alert thresholds

Taken from the client's own recommended alerts.

Band	Condition	Behaviour
Safe	Utilisation below 80%	Proceeds normally
Watch	80% to 90%	Warning shown, transaction permitted
Critical	90% and above	Warning; further commitments need exception approval
Breach	Exposure exceeds approved budget	Hard stop — budget revision or exception approval required

Evidence that shaped the build

The connector is not written from memory. It reads a machine-generated inventory of Zoho's own published OpenAPI specification, so it cannot advertise an endpoint that does not exist. Parsing that specification also settled several questions the documentation website could not.

Findings from Zoho's official specification

ID	Finding	Effect on the build
OAS-01	The ERP API is not uniformly /erp/v3. 14 of 78 modules are served from /erp/v1.	Base path resolves per module, not per connection. Verified live: departments targets /erp/v1/departments.
OAS-02	Purchase Request has no API anywhere in the specification — zero paths, zero modules, zero scopes.	Purchase Request is owned by this application. Custom Modules is the documented fallback carrier.
OAS-03	Purchase-receive lines carry no link to a PO line, no project or WBS coding, and the module has no list endpoint.	Received-but-unbilled is reconstructed here rather than read from Zoho. Stated on the GRN screen.
OAS-04	Purchase order line items do carry project_id, tags, item_custom_fields, account_id and location_id.	The enabling fact. CAPEX code, WBS code and budget head can ride on the PO line.
OAS-06	There is no API to create a custom-field definition — only to set values on a record.	A manual Zoho UI prerequisite. This is a person-blocker before any live sync.
OAS-09	Fixed assets have no link to a source project, PO or bill, and expose no last_modified_time.	Asset sync is full-refresh only; capitalisation is anchored on our own request record.
OAS-07	The published OAuth scope page lists 16 scope families; the specification carries 163 distinct scopes.	Least-privilege scope design is driven from the specification, not the documentation page.

Bill lines are the exception, and it is the one that matters

Bill lines do carry purchaseorder_item_id, described in the specification as linking a bill line to a specific purchase-order item. So the commitment-to-actual leg — the one that prevents double counting — matches at line level and works. It is only the goods-receipt leg that is broken.

What adversarial verification caught

Every research claim was re-checked by a different agent that re-fetched the source and string-matched the quoted evidence. Four catches would have put wrong control logic into this build:

What was claimed	What the source actually says	Consequence avoided
Assigned value is "commitment plus actuals".	SAP publishes two non-equivalent definitions. The authoritative one is a nine-component aggregate.	Building availability control on the compact definition would systematically understate consumption and permit overspend.
Commitment reduction is governed by the goods-receipt indicator.	There is a second independent gate — the goods-receipt-non-valuated flag.	Value falling into neither commitment nor actual. Drove the received-not-billed bucket.
With no capitalisation key, 100% of debits move to the asset under construction.	SAP says all debits "not assigned to CO receivers" — a qualifier that had been dropped.	Modelling CWIP as a total-cost sink and erasing the pre-operative expense write-off path.
Automatic settlement rule generation is unavailable for capital-investment projects.	That sentence does not appear on the cited page. SAP excludes only individual WBS elements assigned to capital investment measures.	Switching settlement-rule generation off wholesale for a project population that is entirely capital-investment.

Across the corpus, 36 of 1,199 quote-checked claims failed string matching — roughly a 3% fabrication-or-drift rate. That figure is the argument for having verified at all.

Contradictions surfaced, not smoothed over

Conflict	Resolution
The client document classifies Purchase Request as a Zoho ERP standard feature; research found no public API.	Both are true. The product feature exists; the API does not. The client document is not refuted.
The client document contains no WBS hierarchy — only project and budget head.	Multi-level WBS comes from the engagement brief, not the client. Every requirement carries a provenance tag.
The document mandates a budget check at Purchase Request, but its formula counts only open POs.	Two approved PRs could clear the same rupee. Logged as an open question with PR reservation available but off by default.

What the application does

Seventeen screens, organised the way a project controller works rather than the way the data is stored. Terminology follows a frozen glossary so an SAP-experienced user meets familiar words.

Area	Screen	What it does
Work	Executive CAPEX Dashboard	Portfolio tiles, budget vs commitment vs actual vs available per project, exception queue
	My Approval Inbox	Everything awaiting a decision, in one list
	Alerts & Exceptions	Budget breaches, over-billed lines, received-but-unbilled ageing
Project control	CAPEX Projects	Project list with exposure and utilisation
	WBS Explorer	Three-level tree table; parents show rolled-up totals of the whole subtree
	Budget Planning Grid	Original, revisions, current approved, commitment, actual, available, exposure % per element
	Budget Availability Check	Simulate a proposed commitment and see the verdict and full position before acting
	Budget Revisions	Versioned revision register with approver, reason, date and approval reference
Procurement	Purchase Requests	Create, budget-check on submission, route to normal or exception approval
	Commitments (PO)	Header and line view with ordered, billed and open commitment; amend, cancel, close
	GRN & Receipts	Receipts including reversals, with the Zoho linkage limitation stated on screen
	Vendor Bills & CWIP	Bills and credit notes with their WBS and budget-head coding
	Commitment Reconciliation	Line-level PO → receipt → bill with position and exception flags
Closure	Capitalisation Workbench	CWIP balance, readiness checks, asset allocation, approval
Integration	Zoho ERP Connector	Connection profile, OAuth, scope validation, test console, event log
	API Inventory	All 869 vendor-documented operations, with the v1/v3 split visible
Governance	Audit Trail	Every action with actor, timestamp, object and detail

The transaction positions the engine handles

All of these are present in the demo dataset and visible on the reconciliation screen, so the behaviour can be inspected rather than taken on trust.

Position	Behaviour
PO approved, not billed	Full value held as open commitment
Partially received, not billed	Commitment retained; received-not-billed reported separately
Partially billed	Commitment reduces by the billed amount; exposure unchanged
Fully billed	Commitment falls to zero; actual carries the full value
Bill exceeds PO	Flagged as an exception with the excess quantified
PO amended upward	Incremental value revalidated against the budget before acceptance
PO amended downward	Commitment released immediately, no revalidation needed
PO cancelled before billing	Entire commitment released; exposure returns to zero

Position	Behaviour
PO closed with residual	Only the unbilled residual is released
Credit note	Reduces actual CWIP and restores availability
GRN reversal	Reduces received quantity and value
Foreign-currency PO	Currency and exchange rate carried on the order
Non-creditable tax, freight	Included in capitalised cost on both order and bill

Governance behaviour worth noting

- An over-budget purchase request cannot be approved without a reason — the API refuses with HTTP 422 and the reason is written to the audit trail.
- Procurement against a closed, capitalised or abandoned WBS element is blocked with a message naming the required reopening approval.
- Capitalisation cannot be approved unless allocations equal the CWIP balance exactly; any remainder must be recorded as a write-off with a reason.
- Capitalisation warns when open commitments or received-but-unbilled value remain, because the asset value would otherwise be understated.
- Approving a budget revision never modifies the original budget line.

Architecture and code

File	Lines	Responsibility
app/backend/domain.py	348	The control engine — formulas, rollup, budget-owner resolution, availability check, line-level reconciliation
app/backend/db.py	550	Schema and demo dataset. Money is integer paise throughout; floats are never used for money
app/backend/main.py	564	FastAPI routes and transaction handling; serves the front end
app/backend/zoho.py	307	Connector, driven entirely by the verified endpoint inventory
app/frontend/app.js	769	Single-page interface: tree table, grids, dialogs, live check
app/frontend/styles.css	242	Design tokens, dense enterprise layout
app/frontend/index.html	58	Application shell
app/run.py	16	Launcher with optional reseed

Design decisions

Decision	Reason
Money as integer paise	Floating point cannot represent currency exactly. Every amount is an integer; formatting happens only at the edge.
Exposure and available are always derived, never stored	A stored balance can drift out of step with the transactions beneath it. These figures are recomputed from source rows on every read.
Bottom-up rollup over the WBS tree	A parent shows the total of itself and every descendant, which is what an SAP-experienced user expects.
SQLite, no external services	The proof of concept must run anywhere with one command. Schema is standard SQL and ports to PostgreSQL unchanged.
No front-end build step	No npm install, no toolchain to break. The interface is served directly.
Connector reads the vendor specification	An endpoint that is not in Zoho's own specification cannot be referenced, so the integration surface cannot be invented.

Data model

Twenty tables covering the full lifecycle. Foreign keys are enforced.

Group	Tables
Organisation	entity, plant, budget_head, app_user
Project structure	project, wbs_element (self-referencing for unlimited depth)
Budget	budget_line (signed, typed: original / supplement / return / transfer), budget_revision
Procurement	purchase_request, purchase_order, po_line, grn, grn_line
Actuals	bill, bill_line
Closure	capitalisation_request, asset_allocation
Integration	zoho_connection, integration_event
Governance	audit_log

The Zoho connector

The connector runs in mock mode: the request that would be issued is constructed from the specification and logged, and only the network exchange is simulated. Going live is a matter of supplying credentials — the request construction is already real.

A connectivity console tests each required module and reports endpoint, method, scope, result, response time, record count, error code and correlation ID. The verified constraints appear as real behaviour, not documentation:

Module	Resolved endpoint	Scope	Observation
departments	/erp/v1/departments	ERP.department.READ	Confirms the version split is handled
purchasereceives	/erp/v3/purchasereceives/{id}	ERP.purchasereceives.READ	By-ID only — no list endpoint exists
vendor-credits	/erp/v3/vendorcredits	ERP.debitnotes.READ	Scope naming is inconsistent in the vendor spec
bills	/erp/v3/bills	ERP.bills.READ	Fails correctly when the scope is withheld

Withholding two scopes during authorisation produces 43 of 45 granted and a named remediation for each missing scope, with the business impact stated — for example, that without bills access commitment never converts to actual.

Running it

```
cd C:\Users\rakes\Fable5\WBS_ZohoERP_POC
python app/run.py # http://127.0.0.1:8000
python app/run.py --reseed # rebuild the demo dataset first
```

Requires Python 3.11 or later with `fastapi` and `uvicorn`. The database is a single file at `app/data/CAPEX.db`.

Demo dataset

Realistic figures for a greenfield manufacturing plant, deliberately reproducing the client document's own budget-head table so the numbers are recognisable.

Project	Current budget	Commitment	Actual CWIP	Available	Utilisation
CAPEX-2026-001 New Production Line	₹1,60,00,000	₹55,50,000	₹52,00,000	₹52,50,000	67.2%
CAPEX-2026-002 Warehouse Expansion	₹80,00,000	₹25,20,000	₹16,80,000	₹38,00,000	52.5%
CAPEX-2026-003 Solar Power Project	₹50,00,000	₹20,00,000	₹11,00,000	₹19,00,000	62.0%
Portfolio	₹2,90,00,000	₹1,00,70,000	₹79,80,000	₹1,09,50,000	62.2%

The dataset deliberately includes an over-budget request routed to exception approval, a partially received but unbilled order, a bill exceeding its purchase order, a cancelled order, an order closed with residual, a credit note, a GRN reversal, a foreign-currency order, an approved supplement, an approved transfer, a pending revision, an abandoned WBS element and a capitalisation case.

What is not production-ready

Stated plainly

This is a proof of concept. It demonstrates that the control loop is correct and buildable; it is not hardened software.

Area	Current state	Needed for production
Authentication	None — a fixed demo user	Single sign-on, session management, multi-factor authentication
Authorisation	Roles modelled and displayed but not enforced	Enforce the 13-role model at API level, with entity and plant scoping
Zoho connection	Mock mode; requests built for real, network hop simulated	Live credentials in a secret manager, real token lifecycle
Secrets	Stored as references, never as values	Cloud secret manager or vault; rotation and revocation procedures
Concurrency	Single-writer SQLite	PostgreSQL with row-level locking on the budget check — two users must not consume the same remaining budget
Data	Demo figures	Opening balance migration for existing CWIP and open commitments
Testing	Manually verified end to end	Automated regression suite, including the concurrency and reconciliation cases

Before a live Zoho connection

Start this conversation now — it blocks the integration and is not a coding task

Zoho ERP has no API that creates a custom-field definition. Someone with Zoho admin access must create the CAPEX code, WBS code and budget head custom fields in the Zoho user interface before the connector can do anything useful. Every subsequent sync depends on it.

- Register an OAuth client and confirm the redirect URI.
- Confirm which Zoho organisation maps to which legal entity — the design supports more than one.
- Grant the least-privilege scope set the connector reports, not a blanket scope.
- Confirm the CWIP general ledger accounts and asset categories against the chart of accounts.
- Decide the Purchase Request approach: owned here, or mirrored into a Zoho custom module.

Open questions for the client

Question	Why it matters
Should an approved Purchase Request reserve budget before a PO exists?	The client document requires a check at PR but its formula counts only open POs, so two PRs could clear the same rupee. Reservation is built and currently off.
Which definition of consumption governs the availability check?	SAP's authoritative definition is broader than the client's formula. Using the narrower one permits overspend relative to SAP behaviour.
Are the two illustrative datasets in the client document the same project?	Sections 6 and 12 share a budget but differ on commitment and actual; they cannot both describe one project at one moment.
What is the approval matrix — amounts, roles, escalation?	The client document names no thresholds. The 13-role model is modelled but the limits are placeholders.

Supporting material

The evidence base sits alongside the code and is not throwaway scaffolding — the connector reads from it at runtime.

Path	Contents
research/00_intake/	Client requirements extracted from the source document, coverage matrix, contradiction register
research/10_lanes/	15 research lanes, 1,527 evidenced claims with source URLs and verbatim quotes
research/20_verified/	Verification verdicts, the parsed OpenAPI registry, and the 869-row endpoint inventory the connector reads
research/30_contracts/	The 12 frozen registries the build follows — formulas, statuses, entities, glossary, tokens, messages
research/40_sections/	Blueprint sections authored so far

research/30_contracts/C5_formulas.json is the load-bearing one. It carries the client invariant with its worked examples, plus the three verification corrections as binding rulings, so the control logic cannot drift.

Blueprint status

The written blueprint that accompanies this build is partially complete: 33 of 51 sections were authored before session limits interrupted the run. The frozen registries are complete, so the remaining sections can be finished without repeating any research or verification.

Recommended next steps

#	Step	Depends on
1	Walk the running application through the full loop with the project controller and finance	Nothing — it runs today
2	Get the Zoho custom fields created in the user interface	Zoho administrator access
3	Confirm the approval matrix and the PR-reservation decision	Client decision
4	Register the OAuth client and connect one organisation for real	Steps 2 and 3
5	Replace SQLite with PostgreSQL and add locking on the budget check	Nothing — can run in parallel
6	Add authentication and enforce the role model	Client identity provider
7	Finish the remaining blueprint sections	Nothing — registries are frozen

A note on scope

The client's document proposed building this control layer inside Zoho ERP. The research does not support that: Zoho ERP projects are flat and carry only a scalar budget, there is no budget-by-head object, no Purchase Request API, and no way to define custom fields programmatically. The multi-level WBS and the budget register have to live in an independent application — which is what has been built — with Zoho remaining the accounting and procurement system of record.

Prepared 6 August 2026. Figures quoted from the running application and from the demo dataset. Zoho capability statements derive from Zoho's own published OpenAPI specification (openapi-all.zip, retrieved 5 August 2026). SAP statements derive from SAP documentation, independently re-fetched and quote-matched during verification. This document describes an independent application; it is not produced, certified or endorsed by SAP or Zoho.